

EXPERIMENT-2:

For a given set of training data examples stored in a .CSV file, implement and demonstrate the Candidate-Elimination algorithm in python to output a description of the set of all hypotheses consistent with the training examples

Code:

```
# =====  
# Candidate Elimination Algorithm  
# Google Colab  
# =====
```

```
import pandas as pd  
from google.colab import files
```

```
# Upload Dataset  
uploaded = files.upload()
```

```
# Read CSV File  
filename = list(uploaded.keys())[0]  
data = pd.read_csv(filename)
```

```
print("Training Dataset:\n")  
print(data)
```

```
# Attributes and Target  
concepts = data.iloc[:, :-1].values  
target = data.iloc[:, -1].values
```

```
# Initialize Specific Hypothesis  
S = concepts[0].copy()
```

```
# Initialize General Hypothesis
```

```
G = [["?" for i in range(len(S))] for j in range(len(S))]
```

```
print("\nInitial Specific Hypothesis:")
```

```
print(S)
```

```
print("\nInitial General Hypothesis:")
```

```
for row in G:
```

```
    print(row)
```

```
# Candidate Elimination Algorithm
```

```
for i, h in enumerate(concepts):
```

```
    if target[i].lower() == "yes":
```

```
        for x in range(len(S)):
```

```
            if h[x] != S[x]:
```

```
                S[x] = "?"
```

```
                G[x][x] = "?"
```

```
    elif target[i].lower() == "no":
```

```
        for x in range(len(S)):
```

```
            if h[x] != S[x]:
```

```
                G[x][x] = S[x]
```

```
            else:
```

```
                G[x][x] = "?"
```

```
print("\nFinal Specific Hypothesis:")
```

```
print(S)
```

```
print("\nFinal General Hypothesis:")
```

```
G_final = []
```

```
for g in G:
```

```
    if g != ["?"] * len(S):
```

```
        G_final.append(g)
```

```
for row in G_final:
```

```
    print(row)
```

Output:

```
... Choose Files EnjoySport (1).csv
EnjoySport (1).csv(text/csv) - 205 bytes, last modified: 8/3/2026 - 100% done
Saving EnjoySport (1).csv to EnjoySport (1).csv
Training Dataset:
```

	Sky	AirTemp	Humidity	Wind	Water	Forecast	EnjoySport
0	Sunny	Warm	Normal	Strong	Warm	Same	Yes
1	Sunny	Warm	High	Strong	Warm	Same	Yes
2	Rainy	Cold	High	Strong	Warm	Change	No
3	Sunny	Warm	High	Strong	Cool	Change	Yes

```
Initial Specific Hypothesis:
['Sunny' 'Warm' 'Normal' 'Strong' 'Warm' 'Same']

Initial General Hypothesis:
['?', '?', '?', '?', '?', '?']
['?', '?', '?', '?', '?', '?']
['?', '?', '?', '?', '?', '?']
['?', '?', '?', '?', '?', '?']
['?', '?', '?', '?', '?', '?']
['?', '?', '?', '?', '?', '?']

Final Specific Hypothesis:
['Sunny' 'Warm' '?' 'Strong' '?' '?']

Final General Hypothesis:
['Sunny', '?', '?', '?', '?', '?']
['?', 'Warm', '?', '?', '?', '?']
```

